Zhiqin "Bill" Qian

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I am a fourth-year PhD student in Computer Science at Rice University, advised by Vaibhav Unhelkar in the Human-AI Teaming Lab. My research focuses on helping AI agents and robots work effectively with people, from turning natural-language instructions into rewards that guide an agent's learning to giving human-robot teams automated feedback after a task.

Education

Rice University, Houston, TX **May 2023 – *present***
Ph.D. in Computer Science, GPA: 4.00/4.00
Advisor: Vaibhav Unhelkar

Rice University, Houston, TX **Aug 2019 – May 2023**
B.A. in Computer Science, Magna Cum Laude, GPA: 3.95/4.00
Distinction in Research and Creative Work
President's Honor Roll (2019, 2021, 2023)

Research Experience

Research Assistant, Human-AI Teaming Lab, Rice University May 2021 – *present*
Building AI agents and robots that work well alongside people.

Research Assistant, Treangen Lab, Rice University May 2020 – Jan 2021
Benchmarked a machine learning tool that scans DNA for harmful pathogens.

Conference Publications

[C2] **Hierarchical Reward Design from Language: Enhancing Alignment of Agent Behavior with Human Specifications**.
Z. Qian, R. Diaz, S. Seo, V. Unhelkar.
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2026.

[C1] **RW4T Dataset: Data of Human-Robot Behavior and Cognitive States in Simulated Disaster Response Tasks**.
L. Orlov-Savko*, Z. Qian*, G. Gremillion, C. Neubauer, J. Canady, V. Unhelkar.
ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2024.

Journal Articles

[J1] **SeqScreen: Accurate and Sensitive Functional Screening of Pathogenic Sequences via Ensemble Learning**.
A. Balaji*, B. Kille*, A. Kappell, G. Godbold, M. Diep, R. Elworth, Z. Qian, D. Albin, D. Nasko, N. Shah, M. Pop, S. Segarra, K. Ternus, T. Treangen.
Genome Biology, 23(1):133, 2022.

Short Papers & Extended Abstracts

[S2] **AI-Assisted Coordination of Human Teams in Situated Tasks**.
S. Seo, Z. Qian, V. Unhelkar.
Proceedings of the AAAI Symposium Series (AAAI Symposium), 2025.

[S1] **Measuring Variations in Workload during Human-Robot Collaboration through Automated After-Action Reviews**.
Z. Qian*, L. Orlov-Savko*, C. Neubauer, G. Gremillion, V. Unhelkar.
Companion of the ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2024.

* equal contribution

Teaching

Teaching Assistant, Rice University

Reinforcement Learning (COMP 552) — Fall 2023, Fall 2025
Guest lectured twice and designed the final exam.
Artificial Intelligence (COMP 557) — Spring 2024
Algorithmic Thinking (COMP 182) — Spring 2021, Spring 2023
Advanced Object-Oriented Programming & Design (COMP 310) — Fall 2022
Reasoning About Algorithms (COMP 382) — Fall 2021

Led office hours and graded homework assignments and exams for all other courses.

Mentoring

Undergraduate Researchers

Arnan Bawa — 2024
Extended the disaster-response simulation with a replay feature.

Isauro Sanchez (Summer Undergraduate Research Fellow) — 2023 – 2024
Built a 3D disaster-response simulation for studying and improving teamwork.

Volunteering

Office of Academic Advising, Rice University — Feb 2020 – May 2023
Peer Academic Advisor
Oversaw and coordinated academic advising programs during orientation week for 100+ new students. Held monthly office hours to help students navigate academic procedures and resources.

Relevant Coursework

Algorithms and Data Structures, Reinforcement Learning, Deep Learning for Vision and Language, Bayesian Statistics, Convex Optimization, Analysis, Linear Algebra, Sciences of the Mind

Last updated: October 28, 2026